

Unifying Special Relativity and Quantum Mechanics

Lecture 11 — The Obstruction, Gauge Invariance, and the Graviton Field

(Ward Identity, Equivalence Classes, Massless Spin-2)

Typed version of the handwritten notes, with added explanations in blue boxes

How to read this document

Conventions. Exactly as in Lectures 1–10: everything in the normal text and in the **yellow / pink** boxes is a faithful transcription of the handwritten notes (the coloured boxes reproduce the results that were framed or highlighted in the original — yellow boxes for the yellow-highlighted results, pink boxes for the pink/red-framed ones). Everything inside a **blue box like this one** is extra material that I added: derivations spelled out, terminology, corrections of transcription slips, and connections to earlier lectures.

Pagination of this scan. The sheets carry the circled numbers 9–29 and continue without interruption the chapter that Lecture 10 opened (the one whose numbering restarted from 1 for the fifth time). They are therefore marked [p. 9****]–[p. 29****], with five stars, exactly as in the second half of Lecture 10.

Overlap with Lecture 10. The first sheet, [p. 9****], is a *repeat*: it is the last page of the Lecture 10 scan. It is reproduced in full in Section 1, because Lecture 11 opens by carrying out precisely the computation it announces.

Where we left off (recap of Lecture 10). We were testing whether a real four-vector field $A^\mu(x)$ can describe massless spin-1 particles. The little group of $p_{\text{REF}} = \kappa \hat{z}$ is $SE(2)$, and the intertwining condition splits into two cases:

- 1a) Λ_W a rotation about $\hat{z}$. This worked: it gave $\varepsilon^0 = \varepsilon^3 = 0$, the two transverse circular polarizations $\varepsilon^\mu(\kappa \hat{z}, \lambda = \pm 1) = \mp \frac{1}{\sqrt{2}}(0, 1, \pm i, 0)$, and the counting $4 - 2 = 2$.
- 1b) Λ_W a pure two-dimensional translation, $\Lambda_W = \exp(-i\alpha T_x - i\beta T_y)$. The Wigner phase vanishes, so the condition reads $\Lambda_W^\mu{}_\nu \varepsilon^\nu(\vec{p}_{\text{REF}}, \lambda) = \varepsilon^\mu(\vec{p}_{\text{REF}}, \lambda)$ — strict invariance.

Lecture 11 begins by computing the left-hand side of that last equation explicitly. It does *not* come out equal to ε^μ — and everything else in this lecture follows from that single failure.

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Part I — The obstruction and gauge invariance of the photon field

1 The invariance condition to be checked [p. 9****]

Consequently the combination

$$H\left(\hat{p}_\Lambda, \frac{|\vec{p}_\Lambda|}{\kappa}\right)^{-1} \Lambda H\left(\hat{p}, \frac{|\vec{p}|}{\kappa}\right) \equiv \Lambda_W(\alpha, \beta).$$

Since in this case $\Lambda_W(\alpha, \beta)$ is a pure two-dimensional translation, the term proportional to J^3 is zero $\Rightarrow \theta(\Lambda_W, p_{\text{REF}}) = 0$.

We get

$$(\Lambda_W(\alpha, \beta))^\mu_\nu \varepsilon^\nu(\vec{p}_{\text{REF}}, \lambda) = \varepsilon^\mu(\vec{p}_{\text{REF}}, \lambda)$$

$\hookrightarrow$ This equation makes a lot of physical sense. Remember that the physical states at reference momentum are defined in such a way that

$$T_x |\vec{p}_{\text{REF}}, \lambda\rangle = T_y |\vec{p}_{\text{REF}}, \lambda\rangle = 0,$$

which imply

$$\exp(-i\alpha T_x - i\beta T_y) |\vec{p}_{\text{REF}}, \lambda\rangle = |\vec{p}_{\text{REF}}, \lambda\rangle.$$

$\hookrightarrow$ This equation, therefore, reflects a property of the physical states.

Let us just check that everything is consistent as it seems to appear... We compute explicitly the left-hand side of this equation:

Added explanation (not in the original notes)

Transcription note. As flagged in Lecture 10, the first factor of the displayed “combination” is written without the inverse in the handwriting; it is restored here. Both outer factors are the identity in this case, so the conclusion is unaffected.

What is about to happen. The equation in the yellow box is a strong demand: the polarization four-vector must be *strictly invariant* under the two little-group translations, not merely invariant up to a phase. The next two pages test it by brute force, and it fails. Keep in mind that step 1a) — the rotation half of $SE(2)$ — caused no trouble at all; it is only the translation half that the four-vector cannot accommodate.

2 The little-group translation as a nilpotent exponential [p. 10****]

$$\Lambda_W(\alpha, \beta) = \exp\left(\underbrace{-i\alpha T_x - i\beta T_y}_{\text{we focus on this combination}}\right)$$

$$\begin{aligned} & -i\alpha(J_{4\text{VEC}}^1 + K_{4\text{VEC}}^2) - i\beta(J_{4\text{VEC}}^2 - K_{4\text{VEC}}^1) = \\ & = -i\alpha \begin{pmatrix} 0 & 0 & i & 0 \\ 0 & 0 & 0 & 0 \\ i & 0 & 0 & -i \\ 0 & 0 & i & 0 \end{pmatrix} - i\beta \begin{pmatrix} 0 & -i & 0 & 0 \\ -i & 0 & 0 & i \\ 0 & 0 & 0 & 0 \\ 0 & -i & 0 & 0 \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & -\beta & \alpha & 0 \\ -\beta & 0 & 0 & \beta \\ \alpha & 0 & 0 & -\alpha \\ 0 & -\beta & \alpha & 0 \end{pmatrix}}_{\equiv N} \end{aligned}$$

“ N ” is a nilpotent matrix of index 3:

$$N^2 = \begin{pmatrix} \alpha^2 + \beta^2 & 0 & 0 & -(\alpha^2 + \beta^2) \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \alpha^2 + \beta^2 & 0 & 0 & -(\alpha^2 + \beta^2) \end{pmatrix}, \quad N^3 = 0.$$

We thus have

► all other terms vanish

$$\Lambda_W(\alpha, \beta) = \exp(N) = \mathbf{1}_{4 \times 4} + N + \frac{1}{2}N^2$$

We thus find

$$\Lambda_W(\alpha, \beta) = \mathbf{1}_{4 \times 4} + \begin{pmatrix} (\alpha^2 + \beta^2)/2 & -\beta & \alpha & -(\alpha^2 + \beta^2)/2 \\ -\beta & 0 & 0 & \beta \\ \alpha & 0 & 0 & -\alpha \\ (\alpha^2 + \beta^2)/2 & -\beta & \alpha & -(\alpha^2 + \beta^2)/2 \end{pmatrix}$$

Added explanation (not in the original notes)

Transcription note (sign). In the handwritten final matrix the third row is written with a minus in the first entry, i.e. $(-\alpha, 0, 0, \dots)$. It should read $(\alpha, 0, 0, -\alpha)$, which is simply the third row of N (the matrix N^2 contributes nothing to that row). The version above is the corrected one; note that the handwritten N itself, and the explicit computation on the next page, both use $(\alpha, 0, 0, -\alpha)$, so this is an isolated slip of the pen and nothing downstream depends on it.

Checking the assembly of N . With i's carried through: $-i\alpha \cdot i = \alpha$ and $-i\beta \cdot (-i) = -\beta$. Row by row, the first matrix contributes $(0, 0, \alpha, 0)$ to row 0, $(\alpha, 0, 0, -\alpha)$ to row 2 and $(0, 0, \alpha, 0)$ to row 3; the second contributes $(0, -\beta, 0, 0)$ to row 0, $(-\beta, 0, 0, \beta)$ to row 1 and $(0, -\beta, 0, 0)$ to row 3. Adding gives N .

Why N is nilpotent, without computing. N is (up to normalization) the matrix representing a *null rotation*. Its structure is governed by the fact that T_x, T_y annihilate the null vector $p_{\text{REF}} = \kappa(1, 0, 0, 1)$ and map the transverse plane into the p_{REF} direction. So N acts as

$$\text{transverse} \longrightarrow p_{\text{REF}}\text{-direction} \longrightarrow 0,$$

a two-step ladder: apply N three times and you necessarily reach zero. That is exactly “nilpotent of index 3”, and it is the reason the exponential *truncates* to a finite polynomial — no approximation is being made in the boxed formula, it is exact.

Reading off the structure of N^2 . Every row of N^2 is proportional to $(1, 0, 0, -1)$, and only rows 0 and 3 are non-zero. Since $p_{\text{REF}}^\mu = \kappa(1, 0, 0, 1)$ has *upper* index while these are rows acting on the lower index, this pattern is the statement that the image of N^2 is one-dimensional and lies along p_{REF} . That will matter on the next page: the polarization vector has vanishing 0 and 3 components, so N^2 annihilates it and only the linear term survives.

3 The explicit computation: the obstruction [p. 11****]

We now compute $\Lambda_W(\alpha, \beta)^\mu{}_\nu \varepsilon^\nu(\vec{p}_{\text{REF}}, \lambda)$ with

$$\varepsilon^\mu(\vec{p}_{\text{REF}}, \lambda) = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ -\lambda \\ -i \\ 0 \end{pmatrix}.$$

$$\begin{aligned}
\Lambda_W(\alpha, \beta)^\mu{}_\nu \varepsilon^\nu(\vec{p}_{\text{REF}}, \lambda) &= \left[\mathbf{1}_{4 \times 4} + \underbrace{N + \frac{1}{2}N^2}_{\text{only “}N\text{” contributes}} \right] \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ -\lambda \\ -i \\ 0 \end{pmatrix} \\
&= \left[\mathbf{1}_{4 \times 4} + \begin{pmatrix} 0 & -\beta & \alpha & 0 \\ -\beta & 0 & 0 & \beta \\ \alpha & 0 & 0 & -\alpha \\ 0 & -\beta & \alpha & 0 \end{pmatrix} \right] \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ -\lambda \\ -i \\ 0 \end{pmatrix} \\
&= \varepsilon^\mu(\vec{p}_{\text{REF}}, \lambda) + \frac{1}{\sqrt{2}} \begin{pmatrix} \beta\lambda - i\alpha \\ 0 \\ 0 \\ \beta\lambda - i\alpha \end{pmatrix} = \varepsilon^\mu(\vec{p}_{\text{REF}}, \lambda) + \frac{(\beta\lambda - i\alpha)}{\sqrt{2}\kappa} \begin{pmatrix} \kappa \\ 0 \\ 0 \\ \kappa \end{pmatrix} \\
&= \varepsilon^\mu(\vec{p}_{\text{REF}}, \lambda) + \frac{(\beta\lambda - i\alpha)}{\sqrt{2}\kappa} p_{\text{REF}}^\mu
\end{aligned}$$

$$\Lambda_W(\alpha, \beta)^\mu{}_\nu \varepsilon^\nu(\vec{p}_{\text{REF}}, \lambda) = \varepsilon^\mu(\vec{p}_{\text{REF}}, \lambda) + \frac{(\beta\lambda - i\alpha)}{\sqrt{2}\kappa} p_{\text{REF}}^\mu \quad \triangle$$

It should be equal to $\varepsilon^\mu(\vec{p}_{\text{REF}}, \lambda) \dots$ but $\beta\lambda - i\alpha \neq 0$ since $\alpha, \beta \in \mathbb{R}!$

Added explanation (not in the original notes)

The two matrix products, line by line. With $\varepsilon = \frac{1}{\sqrt{2}}(0, -\lambda, -i, 0)^T$:

$$(N\varepsilon)^0 = \frac{1}{\sqrt{2}}[(-\beta)(-\lambda) + \alpha(-i)] = \frac{\beta\lambda - i\alpha}{\sqrt{2}}, \quad (N\varepsilon)^3 = \text{the same (row 3 equals row 0),}$$

while rows 1 and 2 of N have non-zero entries only in columns 0 and 3, where ε vanishes — hence $(N\varepsilon)^1 = (N\varepsilon)^2 = 0$. For N^2 : its non-zero columns are 0 and 3, exactly where ε has zeros, so $N^2\varepsilon = 0$. That is the content of the pink annotation “only N contributes”.

Why $\beta\lambda - i\alpha$ cannot vanish. α and β are *real* parameters of the little-group translation and $\lambda = \pm 1$. A complex number $\beta\lambda - i\alpha$ vanishes only if its real and imaginary parts vanish separately, i.e. $\alpha = \beta = 0$ — the identity transformation. For any genuine translation the extra term is non-zero. There is no choice of normalization or phase convention that can rescue this.

The crucial feature: the extra term is along p_{REF}^μ . The failure is not arbitrary. The polarization vector does not become some random new vector — it acquires a multiple of the *momentum itself*. Everything that follows in this lecture is an exploitation of that fact. Notice that it also preserves the transversality condition: $p_{\text{REF}} \cdot p_{\text{REF}} = 0$ for a massless momentum, so $p_{\text{REF}} \cdot (\varepsilon + Cp_{\text{REF}}) = p_{\text{REF}} \cdot \varepsilon = 0$. The new vector is just as good a polarization vector as the old one — which is the first hint that the two might describe the same physics.

4 The contradiction, stated [p. 12****]

We arrive at a **contradiction**.

It is not possible to construct a four-vector quantum field that describes massless spin-1 particles.

The obstruction is in the equation

$$\begin{array}{ccc} & \Lambda_W(\alpha, \beta)^\mu{}_\nu \varepsilon^\nu(\vec{p}_{\text{REF}}, \lambda) & \\ // & & \\ \varepsilon^\mu(\vec{p}_{\text{REF}}, \lambda) & & \backslash\backslash \\ & \varepsilon^\mu(\vec{p}_{\text{REF}}, \lambda) + C_\lambda p_{\text{REF}}^\mu & \end{array} \quad \blacktriangleright \quad C_\lambda \equiv \frac{\lambda\beta - i\alpha}{\sqrt{2}\kappa}$$

On the one hand, $\Lambda_W(\alpha, \beta)\varepsilon$ should be equal to ε in order to reflect the fact that physical states do not transform under little group translations.

We actually have that polarization vectors do transform non trivially under little group translations.

Polarization vectors do not share the same transformation properties of the physical states they are supposed to describe.

Possible solutions?

Added explanation (not in the original notes)

Why this is a genuine crisis and not a technicality. The whole construction of Lectures 8–10 rests on one requirement: the coefficient functions must *intertwine* the finite-dimensional Lorentz representation carried by the field with the infinite-dimensional unitary representation carried by the physical one-particle states. If they do not, the field does not create the states we want it to create, and Lorentz invariance of the S -matrix is lost.

Here the mismatch is sharp. On the state side, $T_x|\vec{p}_{\text{REF}}, \lambda\rangle = T_y|\vec{p}_{\text{REF}}, \lambda\rangle = 0$ was not a convention — it was forced by discarding the continuous-spin representations of $SE(2)$ (Lecture 2). On the field side, the four-vector simply cannot reproduce that invariance. Note that the massive case had no analogous problem: there the little group $SO(3)$ is compact and its representations are genuinely finite-dimensional matrices, which a four-vector can carry. The non-compactness of $SE(2)$, and specifically its translations, is what breaks the correspondence.

5 Two possible solutions; equivalence classes [p. 13****]

(i) We abandon the possibility of describing massless spin-1 particles with a four-vector field.

↪ *Motivation:* the particle–field correspondence is not unique.

↪ However, as we shall discuss, the theory one gets is not compatible with Maxwell electromagnetism.

(ii) We insist with the four-vector description, and go back to the obstruction.

$$\begin{array}{ccc} & \Lambda_W(\alpha, \beta)^\mu{}_\nu \varepsilon^\nu(\vec{p}_{\text{REF}}, \lambda) & \\ // & & \backslash\backslash \\ \varepsilon^\mu(\vec{p}_{\text{REF}}, \lambda) & \stackrel{!}{=} & \varepsilon^\mu(\vec{p}_{\text{REF}}, \lambda) + C_\lambda p_{\text{REF}}^\mu \end{array}$$

↪ Since $\varepsilon^\mu(\vec{p}_{\text{REF}}, \lambda)$ and $\varepsilon^\mu(\vec{p}_{\text{REF}}, \lambda) + C_\lambda p_{\text{REF}}^\mu$ must describe the same physical state, we just impose they do so.

In more technical words, we organize polarization vectors into equivalence classes:

$$\left\{ \varepsilon^\mu(\vec{p}_{\text{REF}}, \lambda) \equiv \varepsilon^\mu(\vec{p}_{\text{REF}}, \lambda) + C_\lambda p_{\text{REF}}^\mu, \quad C_\lambda \in \mathbb{C} \right\}$$

Added explanation (not in the original notes)

What option i) means, and why the notes set it aside. Abandoning the four-vector does not mean abandoning the photon. A massless helicity- ± 1 particle can be described covariantly by the antisymmetric field strength $F^{\mu\nu} \sim (1, 0) \oplus (0, 1)$, which carries no such obstruction: the field strength is gauge invariant to begin with. The price is that the interactions one can write down with $F^{\mu\nu}$ alone are constrained — the minimal coupling $j_\mu A^\mu$ of electrodynamics simply cannot be built, since A^μ does not exist in that formulation. One gets a consistent free theory but not Maxwell's electromagnetism with charged matter. That is exactly the remark “*not compatible with Maxwell electromagnetism*”.

Why the equivalence relation is legitimate. Imposing $\varepsilon \sim \varepsilon + C p_{\text{REF}}^\mu$ is not a trick to sweep the problem away; it is the recognition that the map from polarization vectors to physical states was never injective. The physical state $|\vec{p}_{\text{REF}}, \lambda\rangle$ is one object; the four-vector ε^μ carries four complex components to describe it. The redundancy is the excess baggage, and the little-group translations are what reveal its precise size: exactly one complex direction, along p_{REF}^μ .

Check that the relation is an equivalence relation. Reflexive ($C = 0$), symmetric ($C \rightarrow -C$), transitive ($C_1 + C_2$) — because the shifts form the additive group $\mathbb{C}$. Also note that C_λ is allowed to be *any* complex number in the boxed definition, even though the specific value produced by the little group was $(\lambda\beta - i\alpha)/(\sqrt{2}\kappa)$; as α, β range over $\mathbb{R}$ that expression already sweeps out all of $\mathbb{C}$, so nothing is being added by hand.

6 The equivalence class at generic momentum [p. 14****]

Polarization vectors $\varepsilon^\mu(\vec{p}_{\text{REF}}, \lambda)$ that differ by a complex multiple of the reference momentum must describe the same physical state.

Note that this fact remains true also at generic momentum. Consider the transformation

$$\varepsilon^\mu(\vec{p}, \lambda) = H\left(\hat{p}, \frac{|\vec{p}|}{\kappa}\right)_\nu^\mu \varepsilon^\nu(\vec{p}_{\text{REF}}, \lambda)$$

applied to the equivalence class of reference momentum polarization vectors:

$$\begin{aligned} \varepsilon^\mu(\vec{p}, \lambda) &= H\left(\hat{p}, \frac{|\vec{p}|}{\kappa}\right)_\nu^\mu \left(\varepsilon^\nu(\vec{p}_{\text{REF}}, \lambda) + C_\lambda p_{\text{REF}}^\nu \right) \\ &= \underbrace{H\left(\hat{p}, \frac{|\vec{p}|}{\kappa}\right)_\nu^\mu \varepsilon^\nu(\vec{p}_{\text{REF}}, \lambda)}_{= \varepsilon^\mu(\vec{p}, \lambda)} + C_\lambda \underbrace{H\left(\hat{p}, \frac{|\vec{p}|}{\kappa}\right)_\nu^\mu p_{\text{REF}}^\nu}_{= p^\mu} \\ &= \varepsilon^\mu(\vec{p}, \lambda) + C_\lambda p^\mu \end{aligned}$$

$\varepsilon^\mu(\vec{p}, \lambda) \equiv \varepsilon^\mu(\vec{p}, \lambda) + C_\lambda p^\mu$ must describe the same physical state

$$|\vec{p}, \lambda\rangle = U\left[H\left(\hat{p}, |\vec{p}|/\kappa\right)\right] |\kappa\hat{z}, \lambda\rangle$$

We thus introduce an equivalence class also at non-zero momentum.

Added explanation (not in the original notes)

The one step that does all the work is $H(\hat{p}, |\vec{p}|/\kappa)^\mu_\nu p_{\text{REF}}^\nu = p^\mu$ — which is nothing but the *definition* of the standard boost: it is the Lorentz transformation that carries p_{REF} to p . So the redundancy direction is carried along by the boost and remains the momentum direction at every momentum. The statement is therefore Lorentz covariant even though it was derived at one special momentum, which is precisely what makes it usable.

Consistency with transversality. At generic momentum $p \cdot \varepsilon(\vec{p}, \lambda) = 0$ and $p^2 = 0$, so $p \cdot (\varepsilon + Cp) = 0$: the shifted vector satisfies the same transversality condition. Note also that the shift *changes* ε^0 and $\vec{p} \cdot \vec{\varepsilon}$, both of which were set to zero in Lecture 10 ([p. 6****]–[p. 7****]). This is the resolution of the puzzle noted there: those conditions — the Coulomb gauge — are not Lorentz invariant precisely because they amount to a particular *choice of representative* within the equivalence class, not a physical statement.

7 Consequence 1: the Ward identity [p. 15****–16****]

$$|\vec{p}, \lambda\rangle \longleftrightarrow \left\{ \varepsilon^\mu(\vec{p}, \lambda) \equiv \varepsilon^\mu(\vec{p}, \lambda) + C_\lambda p^\mu, \quad C_\lambda \in \mathbb{C} \right\}$$

Equivalence class of polarization vectors at generic momentum.

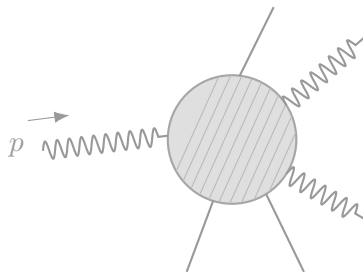
$\hookrightarrow$ We have a redundancy in the field theory description.

Consequences

1) At the level of scattering amplitudes, physics must remain invariant if we use any $\varepsilon^\mu(\vec{p}, \lambda) + C_\lambda p^\mu$ instead of $\varepsilon^\mu(\vec{p}, \lambda)$.

$\hookrightarrow$ In the language of scattering amplitudes, this is known as Ward identity.

Consider a generic scattering amplitude:



The diagram shows a central shaded circle representing a scattering process. Four lines enter and exit the circle: two straight lines and two wavy lines. One wavy line on the left has an arrow pointing towards the circle and is labeled with 'p'. To the right of the diagram, the equation is given: $\equiv i\mathcal{M}(p) = \underbrace{\varepsilon_\mu(\vec{p}, \lambda)}_{\text{incoming photon with momentum } p^\mu} i\mathcal{M}^\mu(p)$.

Since the physics must remain invariant if we consider $\varepsilon^\mu(\vec{p}, \lambda) + C_\lambda p^\mu$ instead of $\varepsilon^\mu(\vec{p}, \lambda)$, we obtain

$$\text{WARD IDENTITY} \quad p_\mu \mathcal{M}^\mu(p) \stackrel{!}{=} 0$$

Added explanation (not in the original notes)

The one-line derivation. Replacing the polarization vector,

$$\mathcal{M} \longrightarrow (\varepsilon_\mu + C_\lambda p_\mu) \mathcal{M}^\mu = \underbrace{\varepsilon_\mu \mathcal{M}^\mu}_{\mathcal{M}} + C_\lambda p_\mu \mathcal{M}^\mu.$$

Demanding that the amplitude be unchanged for every C_λ forces $p_\mu \mathcal{M}^\mu = 0$.

Why this is a strong, testable statement. $\mathcal{M}^\mu$ is computed from Feynman diagrams and knows nothing a priori about polarization vectors. The Ward identity is therefore a non-trivial constraint on the *dynamics*: in QED it is equivalent to conservation of the electromagnetic current, $\partial_\mu j^\mu = 0$, and it survives quantum corrections (where it generalizes to the Ward–Takahashi identities relating the electron self-energy and the vertex function, and is what guarantees that the electron charge is not renormalized independently of the photon field). Practically, it is the standard check on any photon amplitude: replace $\varepsilon_\mu \rightarrow p_\mu$ and the answer must vanish. It also underlies the “ $-g^{\mu\nu}$ ” replacement rule for photon polarization sums in unpolarized cross sections: the extra $p^\mu p^\nu$ terms drop out precisely by the Ward identity.

A remark on the logic. We did not *derive* the Ward identity from a Lagrangian symmetry here. It came from representation theory: the redundancy of the four-vector description of a massless helicity-1 particle. That is the deeper statement — gauge invariance is a *consequence* of insisting on a covariant field for a massless spin-1 particle, not an independent postulate.

8 Consequence 2: gauge invariance of the photon field [p. 16****–17****]

2) The redundancy at the level of polarization vectors reflects into a gauge redundancy at the level of field operators.

Consider

$$\begin{aligned} A^\mu(x) &= \sum_{\lambda=\pm 1} \int \frac{d^3 \vec{p}}{(2\pi)^3 \sqrt{2E_p}} \left[\underbrace{\varepsilon^\mu(\vec{p}, \lambda)}_{\substack{\text{given the equivalence, we substitute} \\ \varepsilon^\mu(\vec{p}, \lambda) \rightarrow \varepsilon^\mu(\vec{p}, \lambda) + C_\lambda p^\mu}} e^{-ip \cdot x} a(\vec{p}, \lambda) + \text{h.c.} \right] \\ &= \sum_{\lambda=\pm 1} \int \frac{d^3 \vec{p}}{(2\pi)^3 \sqrt{2E_p}} \left\{ [\varepsilon^\mu(\vec{p}, \lambda) + C_\lambda p^\mu] e^{-ip \cdot x} a(\vec{p}, \lambda) + \text{h.c.} \right\} \\ &= A^\mu(x) + \sum_{\lambda=\pm 1} \int \frac{d^3 \vec{p}}{(2\pi)^3 \sqrt{2E_p}} \left[\underbrace{C_\lambda p^\mu e^{-ip \cdot x}}_{p^\mu e^{-ip \cdot x} = i \partial^\mu e^{-ip \cdot x}} a(\vec{p}, \lambda) + \text{h.c.} \right] \\ &= A^\mu(x) + \partial^\mu \left\{ \underbrace{\sum_{\lambda=\pm 1} \int \frac{d^3 \vec{p}}{(2\pi)^3 \sqrt{2E_p}} [i C_\lambda e^{-ip \cdot x} a(\vec{p}, \lambda) + \text{h.c.}]}_{\equiv \alpha(x)} \right\} \end{aligned}$$

We thus find

$$A^\mu(x) \equiv A^\mu(x) + \partial^\mu \alpha(x)$$

We thus find a familiar result: the fields $A^\mu(x)$ and $A^\mu(x) + \partial^\mu\alpha(x)$ must describe the same physics, that is the statement of gauge invariance of the photon field.

Added explanation (not in the original notes)

This is the punchline of the whole massless construction. Gauge invariance was not assumed anywhere. The chain was:

$$\begin{aligned} &\text{massless} + \text{spin } 1 + \text{Lorentz covariance} \\ &\Rightarrow \text{four-vector fails little-group translations} \\ &\Rightarrow \varepsilon \sim \varepsilon + Cp \Rightarrow A^\mu \sim A^\mu + \partial^\mu\alpha. \end{aligned}$$

Compare the traditional presentation, in which one *postulates* local $U(1)$ invariance and derives electromagnetism from it. Here the logic runs the other way: the redundancy is forced on us by the representation theory of the Poincaré group, and $U(1)$ gauge invariance is its field-theoretic face.

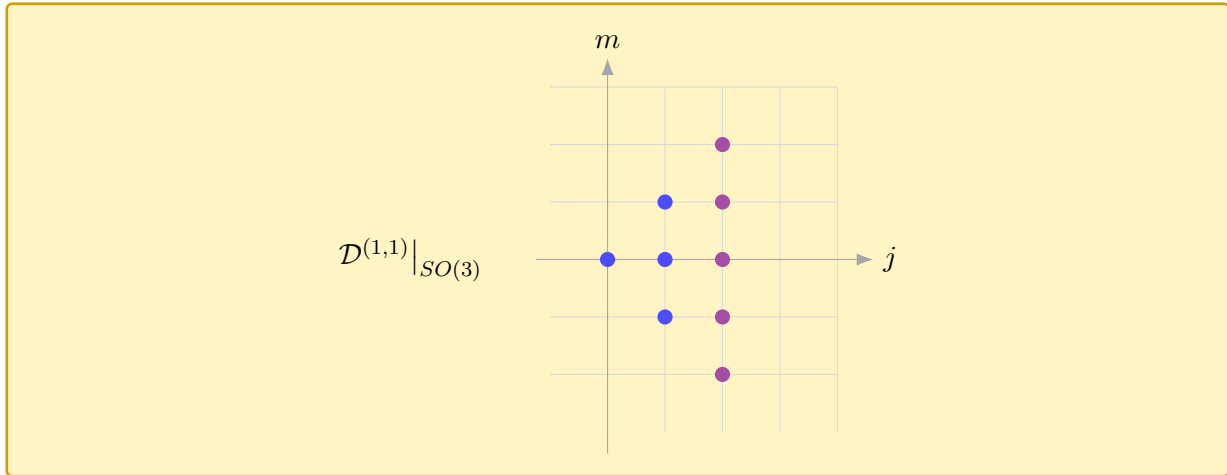
Why $\alpha(x)$ is a legitimate gauge function. Being built from on-shell modes ($p^2 = 0$), it satisfies $\square\alpha = 0$. That is the residual gauge freedom of Lorenz gauge — not the most general α one usually allows, but exactly the part that acts non-trivially on the physical, on-shell content. Nothing is lost: the extra freedom in the general α acts only on off-shell configurations.

Consistency with Consequence 1. The two statements are the same fact seen from two sides. In momentum space $\partial^\mu\alpha \rightarrow ip^\mu\tilde{\alpha}$, so shifting A^μ by a gradient is shifting ε^μ by a multiple of p^μ ; the requirement that amplitudes not notice is the Ward identity.

Part II — The case of the graviton field

9 The case of the graviton field [p. 17****]

We consider the possibility of describing a massless spin-2 particle. We note that



Added explanation (not in the original notes)

Reading the diagram. $\mathcal{D}^{(1,1)}$ restricted to rotations decomposes as $1 \otimes 1 = 0 \oplus 1 \oplus 2$, i.e. $1 + 3 + 5 = 9$ states. The single dot at the origin is the $j = 0$ piece, the column of three at $j = 1$, and the column of five at $j = 2$ (drawn in a different colour in the original). The presence of $j = 2$ is the whole point — by the selection rule of Lecture 8, a representation can describe spin s only if its rotation decomposition contains $j = s$.

Why (1, 1) and not something smaller. The four-dimensional candidates were used up on spin 1. For $s = 2$ the smallest (j_1, j_2) with $j = 2$ in the decomposition is $(1, 1)$, of dimension $(2 \cdot 1 + 1)^2 = 9$. The alternatives $(2, 0)$ and $(0, 2)$ are 5-dimensional and contain $j = 2$ alone, but they are complex and not parity-symmetric — the same reason $(1, 0) \oplus (0, 1)$ rather than $(1, 0)$ describes the field strength. So $(1, 1)$ is the natural choice, exactly as $(1/2, 1/2)$ was for spin 1.

The representation $\mathcal{D}^{(1,1)}$ contains a $j = 2$ component that matches the spin we want to describe $\hookrightarrow$ it seems, therefore, a very natural choice.

10 Rank-2 tensors: a reducible representation [p. 18****]

Let us try to make contact with more “familiar” representations. A rank-2 tensor transforms as the product of two four-vectors:

$$T^{\mu\nu}(x) \xrightarrow{\Lambda} \Lambda^\mu_\rho \Lambda^\nu_\sigma T^{\rho\sigma}(\Lambda^{-1}x)$$

In full generality, $T^{\mu\nu}$ has 16 real independent components.

$\hookrightarrow$ Formally, we write

$$\begin{cases} V^\mu \sim \mathcal{D}^{(1/2, 1/2)} \\ T^{\mu\nu} \sim \underbrace{\mathcal{D}^{(1/2, 1/2)} \otimes \mathcal{D}^{(1/2, 1/2)}} \end{cases}$$

This is a representation of the Lorentz group; however, it is reducible. This means that we should be able to decompose the tensor $T^{\mu\nu}$ into components that transform independently under Lorentz, without mixing into each other.

Added explanation (not in the original notes)

The abstract answer, for orientation. In the (j_1, j_2) language the product decomposes by the usual angular-momentum rule applied to each slot separately:

$$\left(\frac{1}{2}, \frac{1}{2}\right) \otimes \left(\frac{1}{2}, \frac{1}{2}\right) = \left(\frac{1}{2} \otimes \frac{1}{2}, \frac{1}{2} \otimes \frac{1}{2}\right) = (0 \oplus 1, 0 \oplus 1) = (1, 1) \oplus (1, 0) \oplus (0, 1) \oplus (0, 0),$$

with dimensions $9 + 3 + 3 + 1 = 16$. The next three pages find this decomposition concretely, in terms of the symmetric/antisymmetric split and the trace — and the point of doing it concretely is to verify that each piece is genuinely Lorentz-*invariantly* defined.

11 Symmetric and antisymmetric parts [p. 19****]

Let us write

$$T^{\mu\nu} = \underbrace{\frac{1}{2}(T^{\mu\nu} + T^{\nu\mu})}_{\equiv S^{\mu\nu}} + \underbrace{\frac{1}{2}(T^{\mu\nu} - T^{\nu\mu})}_{\equiv A^{\mu\nu}} = S^{\mu\nu} + A^{\mu\nu}$$

$S^{\mu\nu}$ symmetric $A^{\mu\nu}$ antisymmetric

$$T^{\mu\nu} = S^{\mu\nu} + A^{\mu\nu}$$

16 components = 10 components + 6 components

We write a generic rank-2 tensor as the sum of a symmetric and an antisymmetric component.

Importantly, this is a Lorentz-invariant break-up: if we Lorentz-transform a symmetric tensor, it remains symmetric, and if we Lorentz-transform an antisymmetric tensor it remains antisymmetric.

For instance, if we have $S^{\mu\nu} = S^{\nu\mu}$, we write

$$S^{\mu\nu} \longrightarrow S'^{\mu\nu} = \Lambda^\mu_\rho \Lambda^\nu_\sigma S^{\rho\sigma} \overset{\text{using}}{S^{\rho\sigma} \equiv S^{\sigma\rho}} \Lambda^\mu_\rho \Lambda^\nu_\sigma S^{\sigma\rho} \overset{\text{we rename}}{\sigma \leftrightarrow \rho} \Lambda^\mu_\sigma \Lambda^\nu_\rho S^{\rho\sigma} = \Lambda^\nu_\rho \Lambda^\mu_\sigma S^{\rho\sigma} = S'^{\nu\mu}$$

Added explanation (not in the original notes)

Why “Lorentz-invariant break-up” is the key phrase. Splitting a set of components into groups is easy; the content of the statement is that the split *commutes with the group action*. If a Lorentz transformation could turn a symmetric tensor into one with an antisymmetric piece, the two groups would not be separate representations and the decomposition would be meaningless. The three-line computation above is the proof for the symmetric part; the antisymmetric case is identical with one extra sign.

Counting. A 4×4 symmetric matrix has $4(4+1)/2 = 10$ independent entries, an antisymmetric one $4(4-1)/2 = 6$. Note that $6 = 3 + 3$, matching $(1, 0) \oplus (0, 1)$: the antisymmetric rank-2 tensor is exactly the structure of the electromagnetic field strength $F^{\mu\nu}$, whose six components are $\vec{E}$ and $\vec{B}$.

12 Removing the trace [p. 20****]

We now focus on $S^{\mu\nu}$:

$$S^{\mu\nu} = \underbrace{S^{\mu\nu} - \frac{1}{4}g^{\mu\nu}S^\lambda_\lambda}_{\equiv \hat{S}^{\mu\nu}} + \frac{1}{4}g^{\mu\nu}S^\lambda_\lambda = \hat{S}^{\mu\nu} + \frac{1}{4}g^{\mu\nu}S^\lambda_\lambda$$

$$\begin{aligned} \hat{S}^{\mu\nu} &\text{ is traceless by construction;} \\ S^\lambda_\lambda &= \text{tr}(S) = g_{\mu\nu}S^{\mu\nu} \end{aligned}$$

$$\text{tr}(\hat{S}) = \hat{S}^\mu_\mu = S^\mu_\mu - \frac{1}{4}(4)S^\lambda_\lambda = 0$$

$$S^{\mu\nu} = \hat{S}^{\mu\nu} + \frac{1}{4}g^{\mu\nu}S^\lambda_\lambda$$

10 components = 9 components + 1 component

We write a generic symmetric rank-2 tensor as the sum of a traceless symmetric component and a tensor proportional to $g^{\mu\nu}$.

Again, this is a Lorentz-invariant break-up: a traceless symmetric tensor remains traceless symmetric after a Lorentz transformation.

For instance, if we have $\hat{S}^{\mu\nu} = \hat{S}^{\nu\mu}$, $\hat{S}^\mu_\mu = 0$:

$$\begin{aligned} \hat{S}^{\mu\nu} &\rightarrow \hat{S}'^{\mu\nu} = \Lambda^\mu_\rho \Lambda^\nu_\sigma \hat{S}^{\rho\sigma} = \Lambda^\mu_\rho \Lambda^\nu_\sigma \hat{S}^{\sigma\rho} = \Lambda^\nu_\sigma \Lambda^\mu_\rho \hat{S}^{\sigma\rho} = \hat{S}'^{\nu\mu} \quad (\hat{S}^{\mu\nu} \text{ symmetric}) \\ g_{\mu\nu} \hat{S}'^{\mu\nu} &= \underbrace{g_{\mu\nu} \Lambda^\mu_\rho \Lambda^\nu_\sigma}_{g_{\rho\sigma}} \hat{S}^{\rho\sigma} = g_{\rho\sigma} \hat{S}^{\rho\sigma} = \hat{S}^\rho_\rho = 0 \quad \rightsquigarrow \quad \hat{S}'^\mu_\mu = 0 \quad (\hat{S}'^{\mu\nu} \text{ traceless}) \end{aligned}$$

Added explanation (not in the original notes)

Why the coefficient is 1/4. We need $\hat{S}^{\mu\nu} = S^{\mu\nu} - c g^{\mu\nu} S^\lambda_\lambda$ to be traceless. Taking the trace: $S^\lambda_\lambda - c g^\mu_\mu S^\lambda_\lambda = S^\lambda_\lambda(1 - 4c)$, since $g^\mu_\mu = 4$ in four spacetime dimensions. So $c = 1/4$. In D dimensions it would be $1/D$.

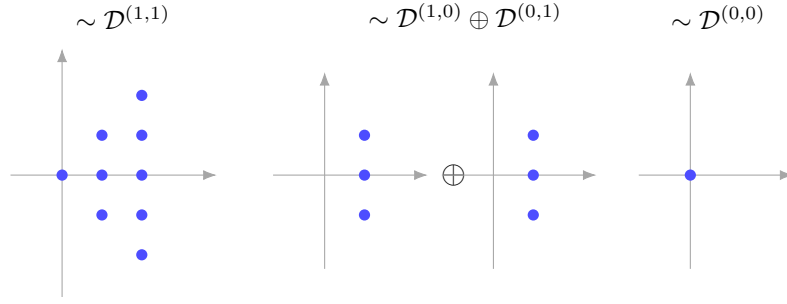
The step that makes the trace part invariant is $g_{\mu\nu} \Lambda^\mu_\rho \Lambda^\nu_\sigma = g_{\rho\sigma}$ — the *defining* property of a Lorentz transformation. This is why the metric can be used to split off a piece covariantly: $g^{\mu\nu}$ is an invariant tensor, the same in every frame. It is the only rank-2 invariant tensor available (besides $\epsilon^{\mu\nu\rho\sigma}$, which has rank 4), which is why the decomposition stops here.

13 The full decomposition: $16 = 9 + 6 + 1$ [p. 21****]

We thus have the following decomposition:

$$16 = 9 + 6 + 1 \quad \text{components}$$

$$T^{\mu\nu} = \hat{S}^{\mu\nu} + A^{\mu\nu} + \frac{1}{4}g^{\mu\nu}S^\lambda_\lambda$$



(Redrawn from the original: weight diagrams in the (j, m) plane of the three irreducible pieces.)

A symmetric traceless rank-2 tensor transforms according to the $(1, 1)$ irrep of the Lorentz group.

$\hookrightarrow$ Since, under rotations, it contains a component with $j = 2$, it is the ideal candidate to describe a spin-2 particle.

We thus introduce a traceless symmetric field

$$h^{\mu\nu}(x) = \sum_{\lambda=\pm 2} \int \frac{d^3\vec{p}}{(2\pi)^3 \sqrt{2E_p}} \left[\varepsilon^{\mu\nu}(\vec{p}, \lambda) e^{-ip \cdot x} a(\vec{p}, \lambda) + \varepsilon^{\mu\nu}(\vec{p}, \lambda)^* e^{ip \cdot x} a^\dagger(\vec{p}, \lambda) \right]$$

with the annotations: we include **both helicities**; $E_p = |\vec{p}|$; $\varepsilon^{\mu\nu}$ is the “**polarization tensor**”, traceless and symmetric.

Added explanation (not in the original notes)

Where each piece of $T^{\mu\nu}$ lands, physically. $A^{\mu\nu}$ (6 components) is the field-strength structure — the Kalb–Ramond / two-form sector. The trace part (1 component) is a Lorentz scalar, i.e. a spin-0 field (the “dilaton” in string-inspired language). The traceless symmetric part (9 components) is the graviton candidate. That the three sectors do not mix under Lorentz is what allows us to keep only the last one.

Why 9 and not 5. A massless spin-2 particle has just *two* helicity states, and even a *massive* one would need only $2s + 1 = 5$. The nine components of $\hat{S}^{\mu\nu}$ are far more than either. As in the spin-1 case, the excess must be removed — and the next pages show that the little group does the job, ending at $9 - 4 - 3 = 2$.

Why $\lambda = \pm 2$ only. For massless particles, helicity is a label of a one-dimensional $SE(2)$ irrep and different $|\lambda|$ values are simply *different particles*. Choosing $\lambda = \pm 2$ is the statement that we are building the graviton; parity invariance requires both signs, exactly as for the photon.

14 Step 1: the polarization tensor at reference momentum [p. 22****–23****]

We look for an explicit expression for the polarization tensors. We follow our strategy.

Step 1: We have the equation

$$(\Lambda_W)^\mu{}_\rho (\Lambda_W)^\nu{}_\sigma \varepsilon^{\rho\sigma}(\vec{p}_{\text{REF}}, \lambda) = e^{-i\theta(\Lambda_W, p_{\text{REF}})\lambda} \varepsilon^{\mu\nu}(\vec{p}_{\text{REF}}, \lambda)$$

$\hookrightarrow$ with $\Lambda_W \in SE(2)$ and $\lambda = \pm 2$

If we take $\Lambda_W = \Lambda_R(\phi, \hat{z})$:

$$\Lambda_R(\phi\hat{z})^\mu{}_\rho \Lambda_R(\phi\hat{z})^\nu{}_\sigma \varepsilon^{\rho\sigma}(\vec{p}_{\text{REF}}, \lambda) = e^{-i\phi\lambda} \varepsilon^{\mu\nu}(\vec{p}_{\text{REF}}, \lambda) \quad (\otimes)$$

with $\lambda = \pm 2$

This equation is solved if we take

$$\varepsilon^{\mu\nu}(\vec{p}_{\text{REF}}, \lambda = \pm 2) = \varepsilon^\mu(\vec{p}_{\text{REF}}, \lambda = \pm 1) \varepsilon^\nu(\vec{p}_{\text{REF}}, \lambda = \pm 1)$$

Individually, these are massless polarization vectors for spin-1, and obey $\Lambda_R(\phi\hat{z})^\mu{}_\nu \varepsilon^\nu(\vec{p}_{\text{REF}}, \lambda) = e^{-i\phi\lambda} \varepsilon^\mu(\vec{p}_{\text{REF}}, \lambda)$ with $\lambda = \pm 1$.

$\hookrightarrow$ It is thus immediate to check that eq. $(\otimes)$ is satisfied.

We thus find the polarization tensor at reference momentum:

$$\varepsilon^{\mu\nu}(\vec{p}_{\text{REF}}, \lambda = \pm 2) = \frac{1}{2} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & \pm i & 0 \\ 0 & \pm i & -1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Added explanation (not in the original notes)

Why the product works. Applying the rotation to each index gives $e^{-i\phi(\pm 1)} \cdot e^{-i\phi(\pm 1)} = e^{-i\phi(\pm 2)}$: the two helicity-1 phases add to give the helicity-2 phase. This is the concrete meaning of “helicities add” and it is why the outer product is the right ansatz.

Deriving the explicit matrix. From Lecture 10 (and [p. 11****] above), $\varepsilon^\mu(\vec{p}_{\text{REF}}, \lambda) = \frac{1}{\sqrt{2}}(0, -\lambda, -i, 0)$. For $\lambda = +1$ this is $\frac{1}{\sqrt{2}}(0, -1, -i, 0)$, so

$$\varepsilon^{11} = \frac{1}{2}, \quad \varepsilon^{12} = \varepsilon^{21} = \frac{(-1)(-i)}{2} = \frac{i}{2}, \quad \varepsilon^{22} = \frac{(-i)^2}{2} = -\frac{1}{2},$$

matching the upper sign. For $\lambda = -1$, $\varepsilon^\mu = \frac{1}{\sqrt{2}}(0, 1, -i, 0)$ and the off-diagonal entry flips to $-i/2$: the lower sign.

Verifying the two defining properties. *Symmetric:* manifestly, since it is $\varepsilon^\mu \varepsilon^\nu$ with the same vector in both slots. *Traceless:* $g_{\mu\nu} \varepsilon^{\mu\nu} = (\varepsilon^0)^2 - (\varepsilon^1)^2 - (\varepsilon^2)^2 = 0 - \frac{1}{2} - (-\frac{1}{2}) = 0$, i.e. $\varepsilon \cdot \varepsilon = 0$ — which holds because the circular polarization vector is a *null* complex vector. Equivalently, the trace of the matrix restricted to the transverse block is $\frac{1}{2}(1 - 1) = 0$.

Why this outer-product structure is special to the massless case. For a *massive* spin-2 particle one would need the full Clebsch–Gordan combination of two spin-1 polarization vectors, since all five m values appear. Here only the extreme helicities $\lambda = \pm 2$ exist, and the extreme-weight state of a tensor product is always the simple product of the extreme-weight states of the factors — no sum over Clebsch–Gordan coefficients is needed.

15 Step 2: generic momentum, transversality, and the counting

[p. 23****–24****]

Step 2: We generate polarization tensors at generic momentum as follows:

$$\varepsilon^{\mu\nu}(\vec{p}, \lambda) = H\left(\hat{p}, \frac{|\vec{p}|}{\kappa}\right)^\mu{}_\rho H\left(\hat{p}, \frac{|\vec{p}|}{\kappa}\right)^\nu{}_\sigma \underbrace{\varepsilon^{\rho\sigma}(\vec{p}_{\text{REF}}, \lambda = \pm 2)}_{\varepsilon^\rho(\vec{p}_{\text{REF}}, \lambda = \pm 1) \varepsilon^\sigma(\vec{p}_{\text{REF}}, \lambda = \pm 1)}$$

$$= \underbrace{\left[H\left(\hat{p}, \frac{|\vec{p}|}{\kappa}\right)^\mu \varepsilon^\rho(\vec{p}_{\text{REF}}, \lambda = \pm 1) \right]}_{\varepsilon^\mu(\vec{p}, \lambda = \pm 1)} \underbrace{\left[H\left(\hat{p}, \frac{|\vec{p}|}{\kappa}\right)^\nu \varepsilon^\sigma(\vec{p}_{\text{REF}}, \lambda = \pm 1) \right]}_{\varepsilon^\nu(\vec{p}, \lambda = \pm 1)}$$

$$\varepsilon^{\mu\nu}(\vec{p}, \lambda = \pm 2) = \varepsilon^\mu(\vec{p}, \lambda = \pm 1) \varepsilon^\nu(\vec{p}, \lambda = \pm 1)$$

polarization tensor at generic momentum

From these expressions, we immediately derive some interesting conclusions. We find, from

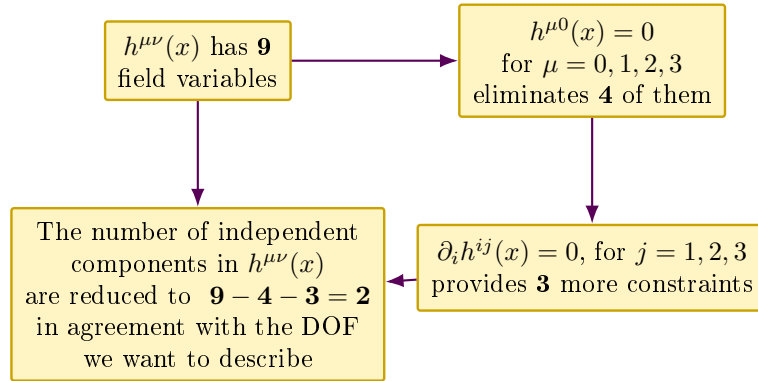
$$\begin{cases} \varepsilon^0(\vec{p}, \lambda = \pm 1) = 0 \\ \vec{p} \cdot \vec{\varepsilon}(\vec{p}, \lambda = \pm 1) = 0 \end{cases}$$

$$\begin{aligned} \varepsilon^{\mu 0}(\vec{p}, \lambda = \pm 2) &= 0 \\ p_i \varepsilon^{ij}(\vec{p}, \lambda = \pm 2) &= 0 \end{aligned}$$

$\Rightarrow$

$$\begin{aligned} h^{\mu 0}(x) &= 0 \\ \partial_i h^{ij}(x) &= 0 \end{aligned}$$

These conditions make sense:



Added explanation (not in the original notes)

Where the two conditions come from. Since $\varepsilon^{\mu\nu} = \varepsilon^\mu \varepsilon^\nu$, any condition on the vector propagates to the tensor. $\varepsilon^0 = 0$ gives $\varepsilon^{\mu 0} = \varepsilon^\mu \varepsilon^0 = 0$ for every μ (and by symmetry $\varepsilon^{0\nu} = 0$). Similarly $p_i \varepsilon^{ij} = (p_i \varepsilon^i) \varepsilon^j = 0$. At the level of the field operator, ∂_i brings down $\mp i p_i$ mode by mode, converting the second into $\partial_i h^{ij} = 0$.

Why 9 and not 10. $h^{\mu\nu}$ is symmetric (10) and traceless (-1) , so 9. The constraint counting then reads: $h^{\mu 0} = 0$ is four conditions ($\mu = 0, 1, 2, 3$), and $\partial_i h^{ij} = 0$ is three ($j = 1, 2, 3$). One might worry about double counting, but $\partial_i h^{i0}$ is already contained in the first set, so the three genuinely new conditions are those with j spatial. Result: $9 - 4 - 3 = 2$, the two graviton helicities.

This is the transverse-traceless (TT) gauge, familiar from general relativity: $h^{0\mu} = 0$, $h^i_i = 0$, $\partial_i h^{ij} = 0$. It is the exact analogue of the Coulomb gauge $A^0 = 0$, $\vec{\nabla} \cdot \vec{A} = 0$ found for the photon in Lecture 10 — and, like it, it is not Lorentz invariant. The reader who noticed the pattern will already suspect what the next page brings.

16 The same problem again: little-group translations [p. 25****]

However, we find the same problem of the photon field. We consider a little group translation, $\Lambda_W(\alpha, \beta)$. We should have

$$\Lambda_W(\alpha, \beta)^\mu{}_\rho \Lambda_W(\alpha, \beta)^\nu{}_\sigma \varepsilon^{\rho\sigma}(\vec{p}_{\text{REF}}, \lambda = \pm 2) = \varepsilon^{\mu\nu}(\vec{p}_{\text{REF}}, \lambda = \pm 2)$$

If we compute explicitly the left-hand side, we get

$$\begin{aligned} & \Lambda_W(\alpha, \beta)^\mu{}_\rho \Lambda_W(\alpha, \beta)^\nu{}_\sigma \varepsilon^{\rho\sigma}(\vec{p}_{\text{REF}}, \lambda = \pm 1) \varepsilon^\sigma(\vec{p}_{\text{REF}}, \lambda = \pm 1) \\ &= \left[\Lambda_W(\alpha, \beta)^\mu{}_\rho \varepsilon^\rho(\vec{p}_{\text{REF}}, \lambda = \pm 1) \right] \left[\Lambda_W(\alpha, \beta)^\nu{}_\sigma \varepsilon^\sigma(\vec{p}_{\text{REF}}, \lambda = \pm 1) \right] \\ &= \left[\varepsilon^\mu(\vec{p}_{\text{REF}}, \lambda = \pm 1) + C_\lambda p_{\text{REF}}^\mu \right] \left[\varepsilon^\nu(\vec{p}_{\text{REF}}, \lambda = \pm 1) + C_\lambda p_{\text{REF}}^\nu \right] \end{aligned}$$

$$\text{with } C_\lambda = \frac{\lambda\beta - i\alpha}{\sqrt{2}\kappa}, \lambda = \pm 1$$

Let us define $C_\pm \equiv \frac{\pm\beta - i\alpha}{\sqrt{2}\kappa}$:

$$\begin{aligned} &= \varepsilon^\mu(\vec{p}_{\text{REF}}, \lambda = \pm 1) \varepsilon^\nu(\vec{p}_{\text{REF}}, \lambda = \pm 1) + p_{\text{REF}}^\mu \left[C_\pm \varepsilon^\nu(\vec{p}_{\text{REF}}, \lambda = \pm 1) + \frac{C_\pm^2}{2} p_{\text{REF}}^\nu \right] \\ &\quad + p_{\text{REF}}^\nu \left[C_\pm \varepsilon^\mu(\vec{p}_{\text{REF}}, \lambda = \pm 1) + \frac{C_\pm^2}{2} p_{\text{REF}}^\mu \right] \end{aligned}$$

If we define

$$N_\pm^\nu(p_{\text{REF}}) \equiv C_\pm \varepsilon^\nu(\vec{p}_{\text{REF}}, \lambda = \pm 1) + \frac{C_\pm^2}{2} p_{\text{REF}}^\nu$$

Added explanation (not in the original notes)

The algebra in one line. Expanding $(\varepsilon^\mu + C p_{\text{REF}}^\mu)(\varepsilon^\nu + C p_{\text{REF}}^\nu)$ gives four terms:

$$\varepsilon^\mu \varepsilon^\nu + C p_{\text{REF}}^\mu \varepsilon^\nu + C p_{\text{REF}}^\nu \varepsilon^\mu + C^2 p_{\text{REF}}^\mu p_{\text{REF}}^\nu.$$

The last term is symmetric in $\mu \leftrightarrow \nu$, so it can be split evenly, $C^2 p_{\text{REF}}^\mu p_{\text{REF}}^\nu = p_{\text{REF}}^\mu \cdot \frac{C^2}{2} p_{\text{REF}}^\nu + p_{\text{REF}}^\nu \cdot \frac{C^2}{2} p_{\text{REF}}^\mu$. Grouping the pieces multiplying p_{REF}^μ and p_{REF}^ν produces exactly $N_\pm^\mu$ and $N_\pm^\nu$. The even split is what makes the final answer manifestly symmetric — as it must be, since $\varepsilon^{\mu\nu}$ is.

Note the structural difference from the photon case. There the shift was $C p_{\text{REF}}^\mu$ — a single term linear in the momentum. Here it is $p_{\text{REF}}^\mu N^\nu + p_{\text{REF}}^\nu N^\mu$ — one shift per index, symmetrized. This is the seed of the fact that the graviton's gauge parameter will be a *vector* ξ^μ , not a scalar.

17 Equivalence classes of polarization tensors [p. 26****]

We find

$$\begin{array}{ccc} \Lambda_W^\mu{}_\rho \Lambda_W^\nu{}_\sigma \varepsilon^{\rho\sigma}(\vec{p}_{\text{REF}}, \lambda = \pm 2) & & \\ \parallel & & \backslash \\ \varepsilon^{\mu\nu}(\vec{p}_{\text{REF}}, \lambda = \pm 2) & & \varepsilon^{\mu\nu}(\vec{p}_{\text{REF}}, \lambda = \pm 2) \\ + p_{\text{REF}}^\mu N_\pm^\nu(p_{\text{REF}}) + p_{\text{REF}}^\nu N_\pm^\mu(p_{\text{REF}}) & & \end{array}$$

We find that polarization tensors do transform non trivially under little group translations.

On the one hand, $\Lambda_W(\alpha, \beta) \Lambda_W(\alpha, \beta) \varepsilon$ should be equal to ε in order to reflect the fact that physical states do not transform under little group translations.

The solution is to introduce an equivalence class of polarization tensors:

$$\left\{ \varepsilon^{\mu\nu}(\vec{p}_{\text{REF}}, \lambda = \pm 2) \equiv \varepsilon^{\mu\nu}(\vec{p}_{\text{REF}}, \lambda = \pm 2) + p_{\text{REF}}^\mu N_\pm^\nu(p_{\text{REF}}) + p_{\text{REF}}^\nu N_\pm^\mu(p_{\text{REF}}) \right\}$$

must describe the same physics.

18 The equivalence class at generic momentum [p. 27****]

We can generalize at generic momentum. Consider the transformation

$$\varepsilon^{\mu\nu}(\vec{p}, \lambda = \pm 2) = H\left(\hat{p}, \frac{|\vec{p}|}{\kappa}\right)^\mu H\left(\hat{p}, \frac{|\vec{p}|}{\kappa}\right)^\nu \varepsilon^{\rho\sigma}(\vec{p}_{\text{REF}}, \lambda = \pm 2)$$

applied to the equivalence class of reference momentum polarization tensors:

$$\begin{aligned} \varepsilon^{\mu\nu}(\vec{p}, \lambda = \pm 2) &= H^\mu_\rho H^\nu_\sigma \left(\varepsilon^{\rho\sigma}(\vec{p}_{\text{REF}}, \lambda) + p_{\text{REF}}^\rho N_\pm^\sigma(p_{\text{REF}}) + p_{\text{REF}}^\sigma N_\pm^\rho(p_{\text{REF}}) \right) \\ &= \varepsilon^{\mu\nu}(\vec{p}, \lambda) + \underbrace{\left[H^\mu_\rho p_{\text{REF}}^\rho H^\nu_\sigma N_\pm^\sigma(p_{\text{REF}}) \right]}_{p^\mu} + \underbrace{\left[H^\nu_\sigma p_{\text{REF}}^\sigma H^\mu_\rho N_\pm^\rho(p_{\text{REF}}) \right]}_{p^\nu} \end{aligned}$$

with

$$\begin{aligned} H^\nu_\sigma \left(C_\pm \varepsilon^\sigma(p_{\text{REF}}, \lambda = \pm 1) + \frac{C_\pm^2}{2} p_{\text{REF}}^\sigma \right) &= C_\pm \varepsilon^\nu(\vec{p}, \lambda = \pm 1) + \frac{C_\pm^2}{2} p^\nu \equiv N_\pm^\nu(p) \\ &= \varepsilon^{\mu\nu}(\vec{p}, \lambda) + p^\mu N_\pm^\nu(p) + p^\nu N_\pm^\mu(p) \end{aligned}$$

We introduce an equivalence class at generic momentum.

$$|\vec{p}, \lambda\rangle \longleftrightarrow \left\{ \varepsilon^{\mu\nu}(\vec{p}, \lambda) \equiv \varepsilon^{\mu\nu}(\vec{p}, \lambda) + p^\mu N_\pm^\nu(p) + p^\nu N_\pm^\mu(p) \right\}$$

Equivalence class of polarization tensors at generic momentum.

Added explanation (not in the original notes)

Same mechanism as for the photon. The standard boost carries $p_{\text{REF}} \rightarrow p$ and, applied to the vector $N_\pm$, produces the corresponding object at generic momentum. So the shift keeps its form, with $p_{\text{REF}} \rightarrow p$ everywhere. Once again the crucial input is $H^\mu_\rho p_{\text{REF}}^\rho = p^\mu$.

The shape of the redundancy. For the photon: $\varepsilon^\mu \rightarrow \varepsilon^\mu + C p^\mu$ — one complex parameter. For the graviton: $\varepsilon^{\mu\nu} \rightarrow \varepsilon^{\mu\nu} + p^\mu N^\nu + p^\nu N^\mu$ — a whole vector N^ν . That extra structure is exactly what will make the gauge parameter of the graviton a vector field.

Consistency check on tracelessness. The shifted tensor should still be traceless if the correspondence is to make sense. Indeed $g_{\mu\nu}(p^\mu N^\nu + p^\nu N^\mu) = 2 p \cdot N$, and using $p^2 = 0$ and $p \cdot \varepsilon(\vec{p}, \lambda = \pm 1) = 0$,

$$p \cdot N_\pm = C_\pm \underbrace{p \cdot \varepsilon}_0 + \frac{C_\pm^2}{2} \underbrace{p^2}_0 = 0.$$

So the shift preserves both symmetry and tracelessness — the equivalence class stays inside the (1,1) representation, as it must. (*This check is not in the notes; it is worth doing once.*)

19 Gauge redundancy of the graviton field [p. 28****–29****]

Let us consider the consequence for the quantum field $h^{\mu\nu}(x)$.

↪ The redundancy in the description of the polarization tensors reflects into a redundancy at the level of field operators.

$$\begin{aligned}
 h^{\mu\nu}(x) &= \sum_{\lambda} \int \frac{d^3\vec{p}}{(2\pi)^3 \sqrt{2E_p}} \left[\underbrace{\varepsilon^{\mu\nu}(\vec{p}, \lambda)}_{\text{given the physical equivalence, we substitute}} e^{-ip \cdot x} a(\vec{p}, \lambda) + \text{h.c.} \right] \\
 &= \sum_{\lambda} \int \frac{d^3\vec{p}}{(2\pi)^3 \sqrt{2E_p}} \left\{ \left[\varepsilon^{\mu\nu}(\vec{p}, \lambda) + p^{\mu} N_{\pm}^{\nu}(p) + p^{\nu} N_{\pm}^{\mu}(p) \right] e^{-ip \cdot x} a(\vec{p}, \lambda) + \text{h.c.} \right\} \\
 &= h^{\mu\nu}(x) + \left\{ \sum_{\lambda} \int \frac{d^3\vec{p}}{(2\pi)^3 \sqrt{2E_p}} \left[\underbrace{p^{\mu} N_{\pm}^{\nu}(p) e^{-ip \cdot x}}_{\text{we use } p^{\mu} e^{-ip \cdot x} = i\partial^{\mu} e^{-ip \cdot x}} a(\vec{p}, \lambda) + \text{h.c.} \right] + (\mu \leftrightarrow \nu) \right\} \\
 &= h^{\mu\nu}(x) + \underbrace{\left\{ \partial^{\mu} i \sum_{\lambda} \int \frac{d^3\vec{p}}{(2\pi)^3 \sqrt{2E_p}} \left[N_{\pm}^{\nu}(p) e^{-ip \cdot x} a(\vec{p}, \lambda) - \text{h.c.} \right] + (\mu \leftrightarrow \nu) \right\}}_{\equiv \xi^{\nu}(x)}
 \end{aligned}$$

We thus find

$$h^{\mu\nu}(x) \equiv h^{\mu\nu}(x) + \partial^{\mu} \xi^{\nu}(x) + \partial^{\nu} \xi^{\mu}(x)$$

The fields $h^{\mu\nu}(x)$ and $h^{\mu\nu}(x) + \partial^{\mu} \xi^{\nu}(x) + \partial^{\nu} \xi^{\mu}(x)$ must describe the same physics.

Added explanation (not in the original notes)

This is linearized general relativity. The transformation

$$h^{\mu\nu} \longrightarrow h^{\mu\nu} + \partial^{\mu} \xi^{\nu} + \partial^{\nu} \xi^{\mu}$$

is exactly the linearized *diffeomorphism* (coordinate change $x^{\mu} \rightarrow x^{\mu} - \xi^{\mu}$) under which the metric perturbation $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ is invariant to first order. So the same argument that produced $U(1)$ gauge invariance for the photon produces *general covariance* for the graviton — again without assuming it. A massless spin-2 particle described by a covariant tensor field *necessarily* comes with diffeomorphism invariance.

The parallel, side by side.

	Photon ($\lambda = \pm 1$)	Graviton ($\lambda = \pm 2$)
Representation	$\mathcal{D}^{(1/2, 1/2)}, A^\mu$	$\mathcal{D}^{(1, 1)}, h^{\mu\nu}$ traceless sym.
Components	4	9
Constraints	$A^0 = 0, \vec{\nabla} \cdot \vec{A} = 0$	$h^{\mu 0} = 0, \partial_i h^{ij} = 0$
Counting	$4 - 2 = 2$	$9 - 4 - 3 = 2$
Polarization shift	$\varepsilon^\mu + C p^\mu$	$\varepsilon^{\mu\nu} + p^\mu N^\nu + p^\nu N^\mu$
Gauge parameter	scalar $\alpha(x)$	vector $\xi^\mu(x)$
Field redundancy	$A^\mu + \partial^\mu \alpha$	$h^{\mu\nu} + \partial^\mu \xi^\nu + \partial^\nu \xi^\mu$
Amplitude identity	$p_\mu \mathcal{M}^\mu = 0$	$p_\mu \mathcal{M}^{\mu\nu} = 0$

The Ward identity for gravity. By the same argument as in Part I, replacing $\varepsilon_{\mu\nu} \rightarrow \varepsilon_{\mu\nu} + p_\mu N_\nu + p_\nu N_\mu$ in an amplitude $\mathcal{M} = \varepsilon_{\mu\nu} \mathcal{M}^{\mu\nu}$ must leave it unchanged, which requires $p_\mu \mathcal{M}^{\mu\nu} = 0$. In gravity this is the statement of energy–momentum conservation, $\partial_\mu T^{\mu\nu} = 0$ — the analogue of current conservation in QED. It is also, famously, what forces the graviton to couple *universally* to all forms of energy with the same strength: the equivalence principle emerges from consistency rather than being postulated.

Where this stops. The construction here is entirely at the level of the free field. The interacting theory is far more constrained: demanding that the gauge redundancy survive interactions order by order forces the full non-linear structure of Einstein’s equations. Note also that no consistent interacting massless field with $\lambda \geq 5/2$ exists in flat space (Weinberg–Witten and related no-go theorems) — so the ladder that ran spin 0, 1/2, 1, 2 stops here.